

University of Dhaka
Department of Computer Science & Engineering
CSE 2216 - Application Development Lab (Batch 28)

Assignment 1 - Project Requirement Analysis

Team Name: Binary

Roll Numbers:

11

26

44

60

Submitted On

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Idea

Our educational platform, designed to enhance learning experience! Our app allows students to browse and enroll in a variety of courses through a secure payment gateway. Once enrolled, students can interact with peers in the same course through chat, fostering a collaborative learning environment. Instructors can easily share Zoom links for live classes, keeping everyone connected and share necessary course materials.. Each course also includes a leaderboard, allowing students to track their progress, and a dedicated Q&A section for discussing course content and questions.

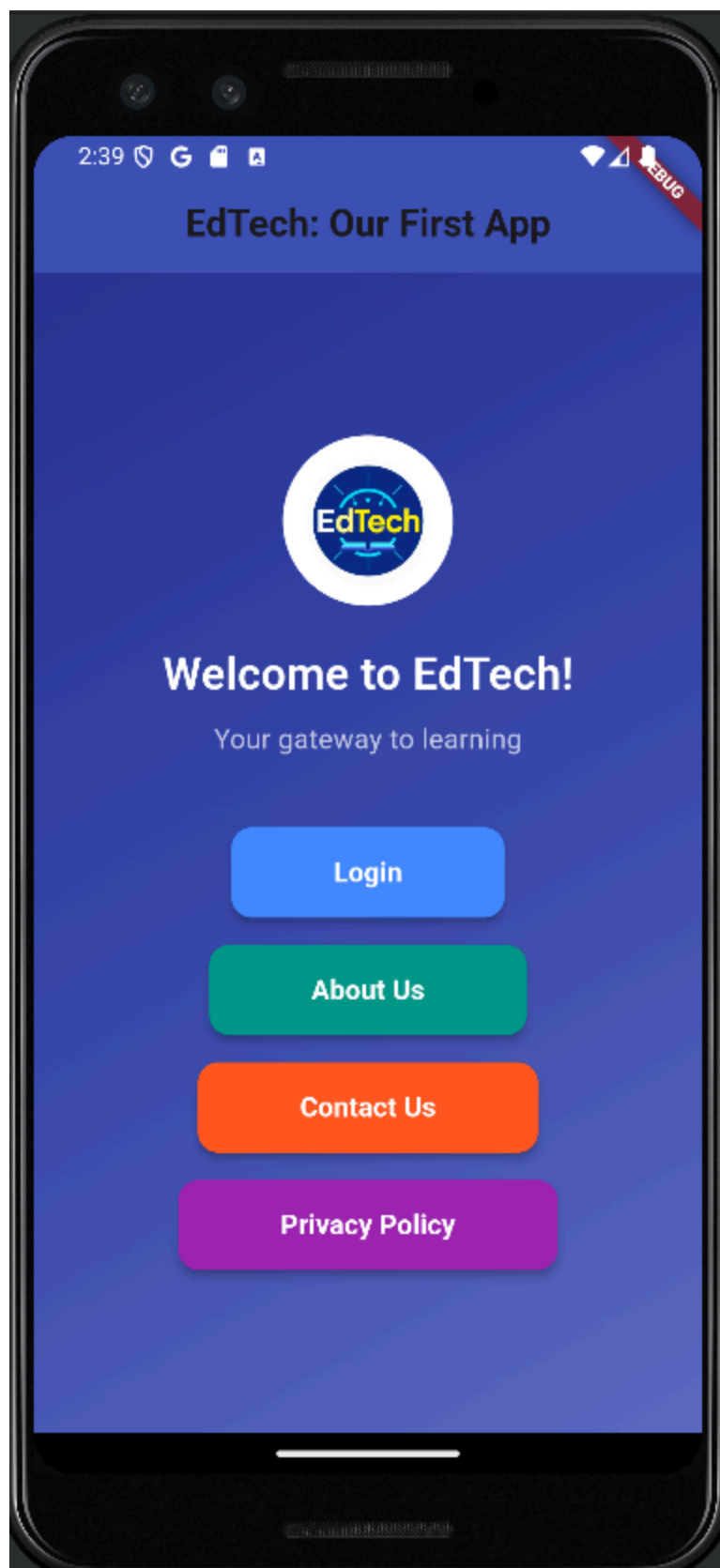
Refer to Existing Applications

1. **10 Minute School** (Local): Bangladeshi e-learning platform that provides academic courses, skill-based lessons, and live classes. Known for its user-friendly interface, accessible content, and course variety.
2. **Programming Hero** (Local): This app aimed at teaching coding in a fun, interactive way, with gamified lessons and a strong community. It allows users to learn programming from scratch with engaging tutorials.
3. **Udemy** (International): An extensive online course marketplace, known for its variety of courses, affordable pricing, and strong instructor-student interaction through Q&A forums and community discussions.
4. **Chorcha** (Local): Chorcha is an educational app widely used for HSC, university admission, and BCS exam preparation. It offers a digital question bank, subject-specific exams, and unlimited practice tests. With features like a leaderboard and model tests, it encourages a competitive learning environment, helping students track their progress and prepare effectively.

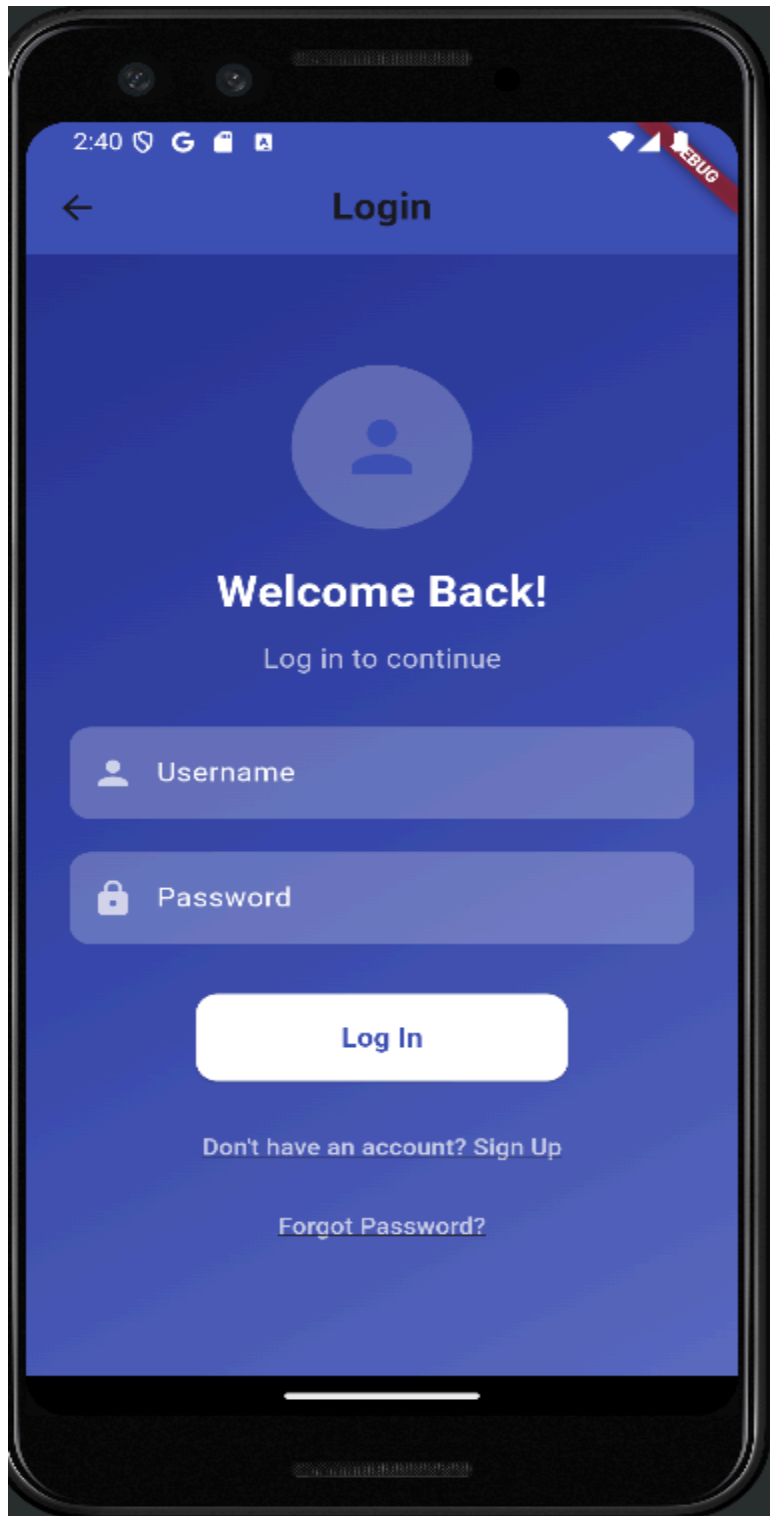
App Screens

- **Opening Screen:** In the opening screen there will be select options for students and teachers. This screen will direct them to their respective login page.
- **Teachers login page:** In this screen there will be a login option and a signup option for new teachers.
- **Teachers signup page:** In this screen teachers can make new accounts. Teachers can select their interested field.
- **Teachers home page:** At the top of this screen there will be a search bar for the existing courses in the application. Then there will be a scrollable section showing all the courses of our platform. At the bottom there will be a teachers account section page link, new course launch page link, existing courses page link.
- **Teacher account settings page:** Here teachers can update their profile information, change passwords, names, contact details and other personal information.
- **Teacher Existing course page:** There will be courses launched by the teacher. Every link will be directed to a course page that is published by the teacher.
- **Teacher New Course Launch page:** Here the teacher can publish new courses and make other necessary requirements for the course like pricing, topic of the course etc.
- **Teacher Specific Course page:** When a teacher enters one of his courses he can schedule class, launch a new class link, reschedule class, launch a quiz, see student statistics, details of students and answer the questions of the students through Q/A section, provide new course material etc.
- **Students login page:** In this screen there will be a login option and a signup option for new students.
- **Student Signup page:** In this screen students can make new accounts. Students can select their interested field.

- **Students home page:** At the top of this screen there will be a search bar for the existing courses in the application. Students can search courses with tags and they can see the details of the course, view teachers, other courses, review of the course etc. Then there will be a scrollable section showing all the courses of our platform. A student can also select courses from this page. At the bottom there will be a students account section page link, existing enrolled courses page link.
- **Student account settings page:** Here students can update their profile information, change passwords, names, contact details and other personal information.
- **Students Enrolled course page:** There will be courses enrolled by the students. Every link will be directed to a course page that is enrolled by the student.
- **Students specific Course page:** When a student enters one of his courses he can view the class materials, class link, schedule, quizzes, leaderboard for the exams, and other teammates, progress report of students.
- **Forgot Password:** Students and teachers both can change their passwords when they forgot.
- **Privacy Policy:** Course copyright, phone access and other necessary parts will be added to this page.
- **Chatting, Review and Commenting section:** A student can review the course he enrolled and add comments in the course that will be answered by the students or teacher. We will try to add chatting features between the coursemates.
- **Live class:** We will **try** to make an inbuilt app zoom live class.
- **Payment gateway:** Students can purchase courses using bkaash.
- **Class Routine:** There will be a page where we will show the online class schedule, exam schedule and other routine related parts.
- **Onboarding screen(Not final):**



- User Login(Not final):



Identify User Persona

1. Primary User Groups

- **Students:** Those are interested in self-paced or guided learning, seek access to online educational resources, peer collaboration, and structured courses with interactive elements.
- **Teachers (Instructors):** Experienced in a particular field and interested in delivering online courses.

2. Demographics

- **Age Range:** Likely 18-35, targeting students and instructors in educational sectors.
- **Location:** Bangladesh, as we are using a local payment gateway system. It may be updated later.

3. User Goals

- **Students:**
 - Access high-quality educational content and courses on-demand.
 - Communicate with peers and instructors for collaborative learning.
 - Track and measure progress within courses.
 - Find a convenient method to pay for courses.
- **Teachers:**
 - Create, manage, and deliver courses efficiently.
 - Connect with students in a streamlined, user-friendly environment.
 - Foster engagement through live classes, quizzes, and discussion features.
 - Manage administrative aspects like class schedules, material uploads, and Q&A.

4. User Needs

- **Students:**

- Easy navigation for discovering and enrolling in courses.
- Direct access to course materials, live class links, schedules, and quizzes.
- Ability to communicate and collaborate with peers.
- Personal progress tracking and a leaderboard for motivation.

- **Teachers:**

- Simple process for launching and managing new courses.
- Tools for scheduling live classes, sharing materials, and assessing student performance.
- Ability to answer student questions and facilitate discussions in a dedicated space.

5. Behaviors

- **Students:**

- Engage with course materials asynchronously but attend live sessions as scheduled.
- Rely on leaderboards and progress tracking for motivation.
- Actively participate in chats, Q&A, and peer discussions to clarify doubts.

- **Teachers:**

- Frequently update course content, class schedules, and quiz materials.
- Respond to students' questions and comments to maintain engagement.
- Use the app as a primary tool for handling administrative tasks and course management.

6. Pain Points

- **Students:**

- Limited access to interactive elements like chat and real-time Q&A in other apps. (We will try a better one though)
- **Teachers:**
 - Overwhelming task of managing multiple administrative and interactive elements for a course.
 - Limited options in other platforms for engaging students beyond simple content delivery.
 - Desire for a streamlined, one-stop platform for live sessions, materials, and course management.

Design Survey

Survey link: [survey link](#)

Survey result link: [result](#)

GitHub Repository

Github Link: [Click here to view github repository](#)